

Demo: Stored Procedures

Summary

Created a stored procedure named `update_sales_price` using PL/pgSQL in Amazon Redshift, used it to update the `pricepaid` value in the `sales` table, handled invalid sales IDs with validation messages, and successfully verified the updated sales record using SQL queries.

Environment Used

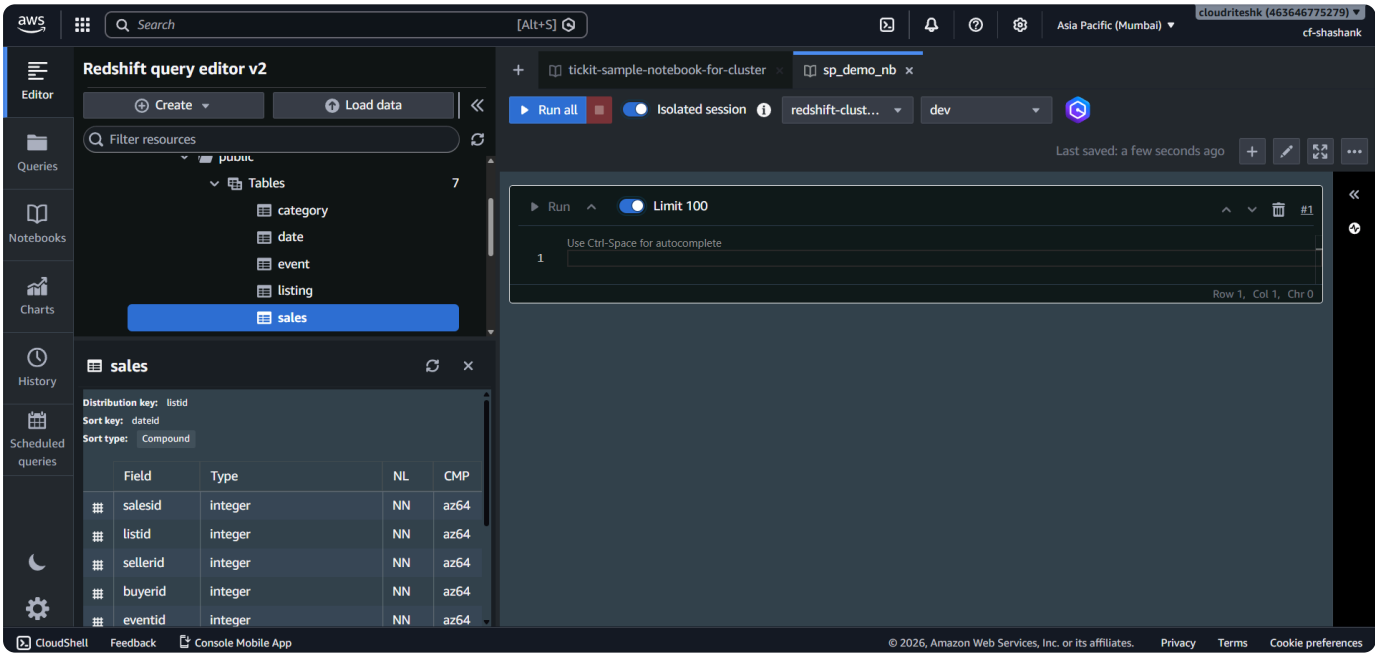
Worked inside:

- Redshift Query Editor v2
- Connected sample Redshift cluster

Database used: `dev`

Schema used: `public`

Exploring the Sales Table



Opened:

dev → public → sales

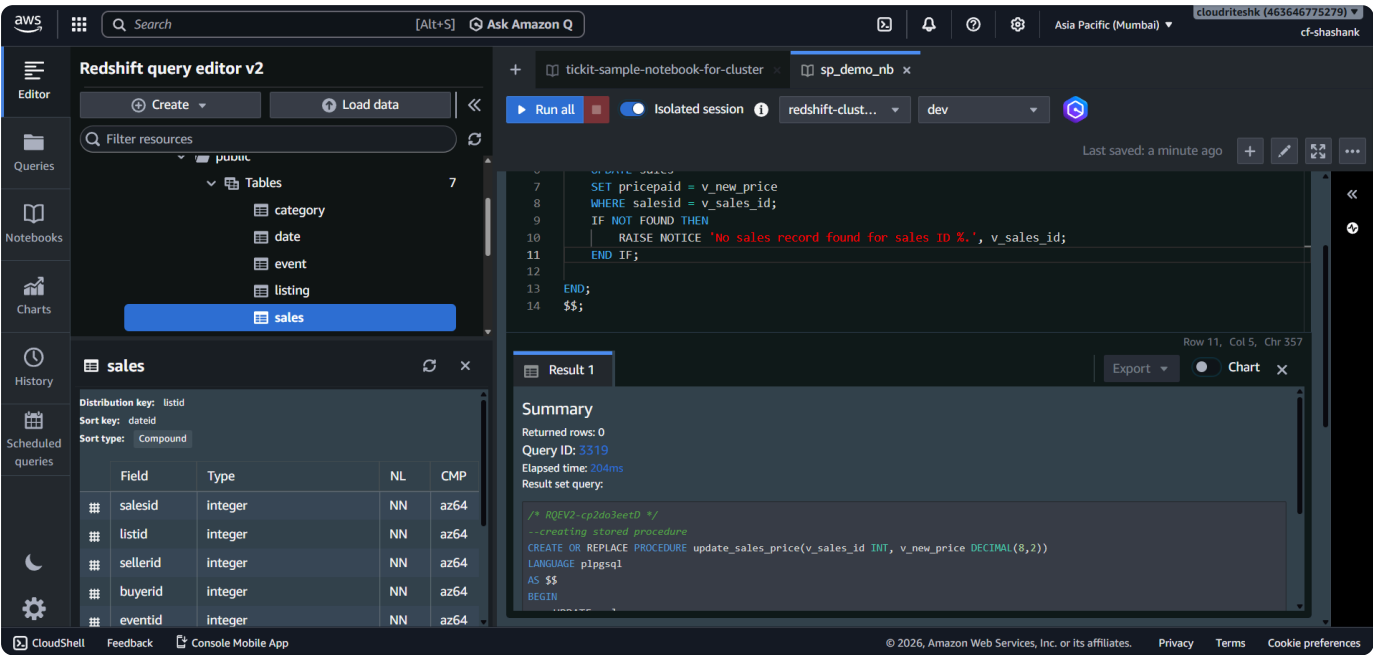
Important columns observed:

sales table

Column	Purpose
salesid	Unique sales record ID
pricepaid	Amount paid for the sale

Creating the Stored Procedure

Purpose



Create a procedure that:

- Takes a sales ID
- Updates the `pricepaid` value for that record

Parameters Used

Parameter	Meaning
<code>v_sales_id</code>	Sales ID to update
<code>v_new_price</code>	New price value

Procedure Command

Used:

```
CREATE OR REPLACE PROCEDURE
```

Language: `plpgsql`

What is PL/pgSQL?

PostgreSQL procedural language used for:

- Conditions
- Loops
- SQL processing logic

Procedure Logic

Inside: `BEGIN`

The procedure:

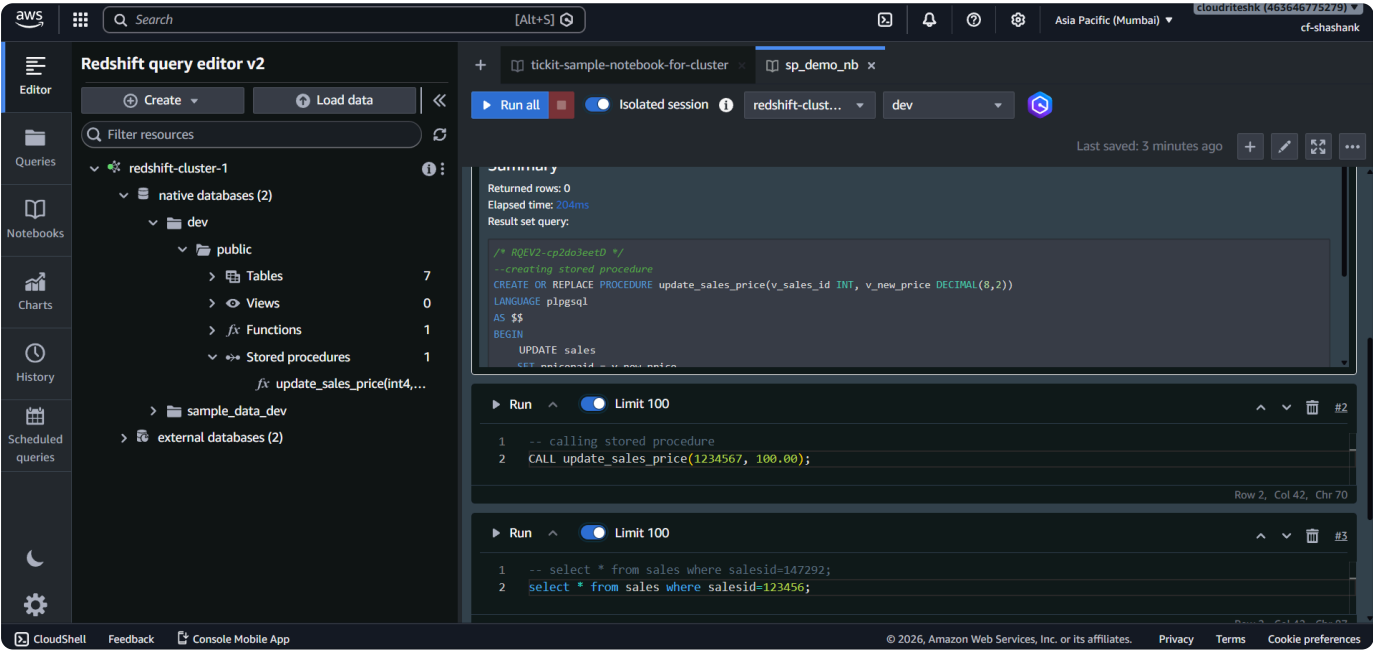
1. Updates the `sales` table
2. Sets: `pricepaid = v_new_price`
3. Uses condition: `salesid = v_sales_id`

Validation Logic

If no matching sales record exists:

- Procedure raises notice: `No sales record found for provided sales ID`

Procedure Creation Result



Initially: `Stored Procedures: 0`

After running the procedure:

- Creation succeeded successfully.

Refreshing the left panel showed: `Stored Procedures: 1`

Visible procedure: `update_sales_price`

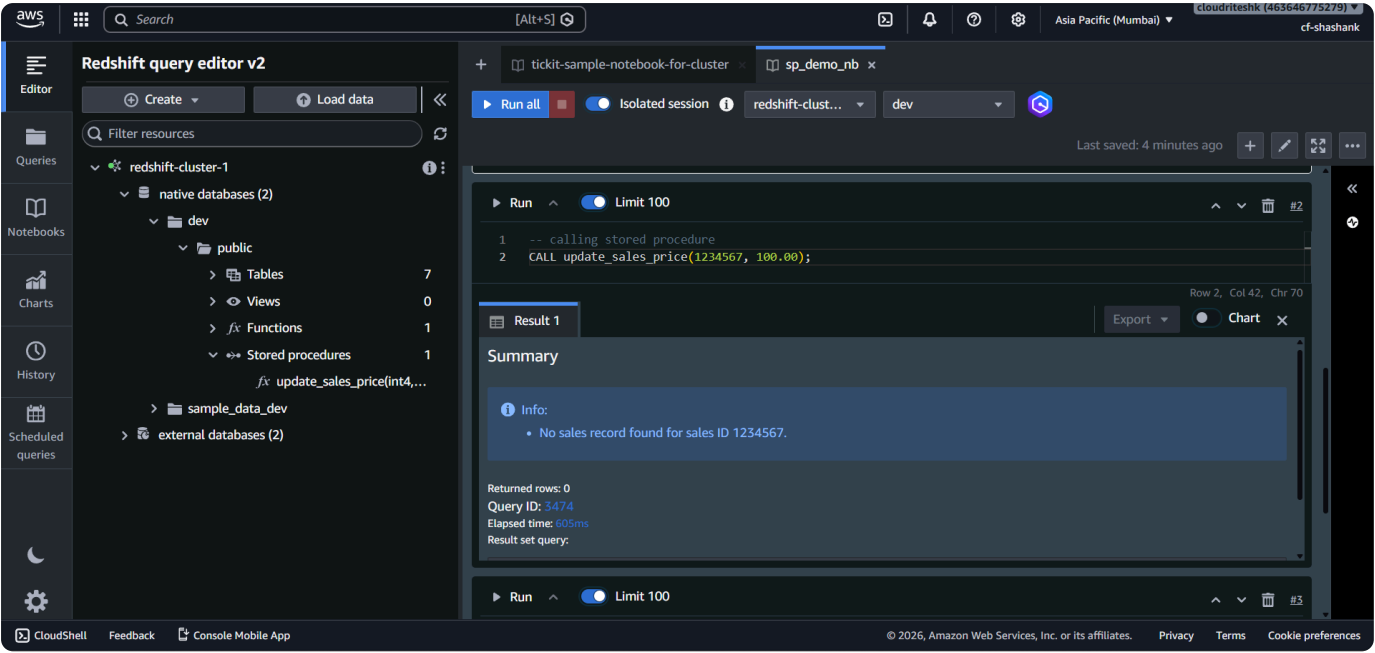
Executing the Stored Procedure

CALL Command Used

Executed procedure using:

```
CALL update_sales_price(...)
```

Test 1 – Invalid Sales ID



Used: 1234567

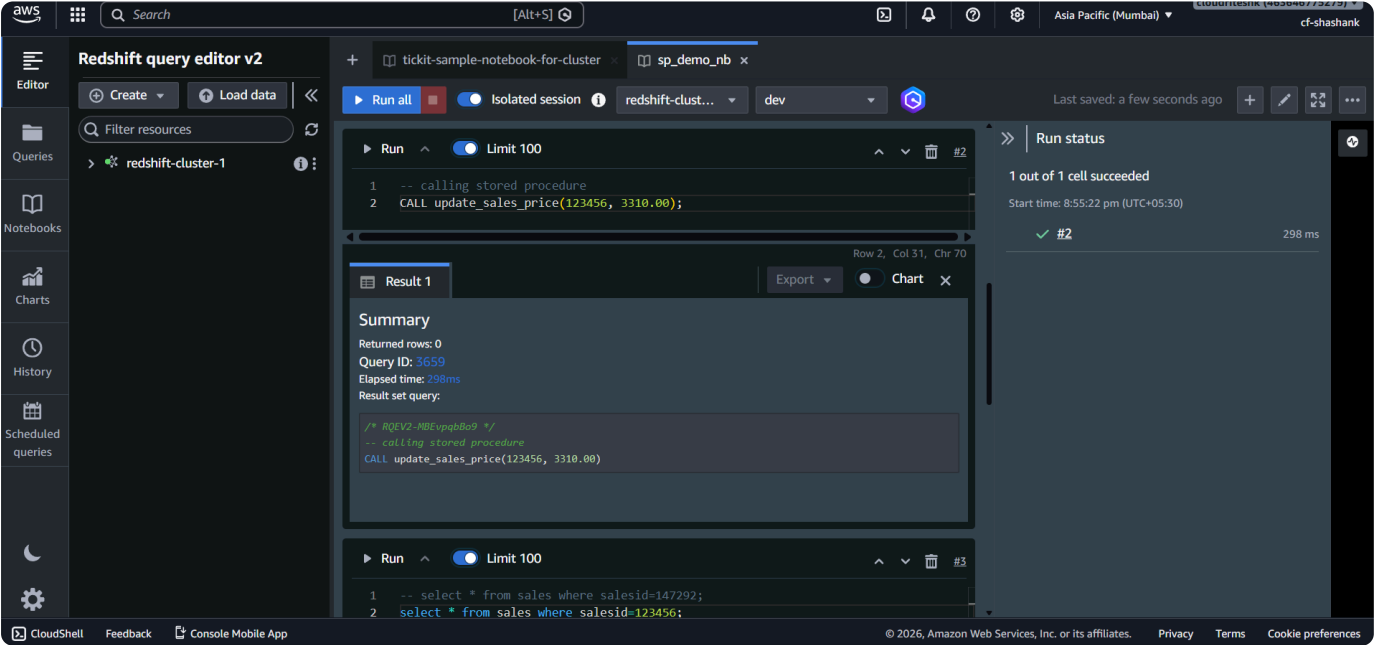
This sales ID did not exist in the table.

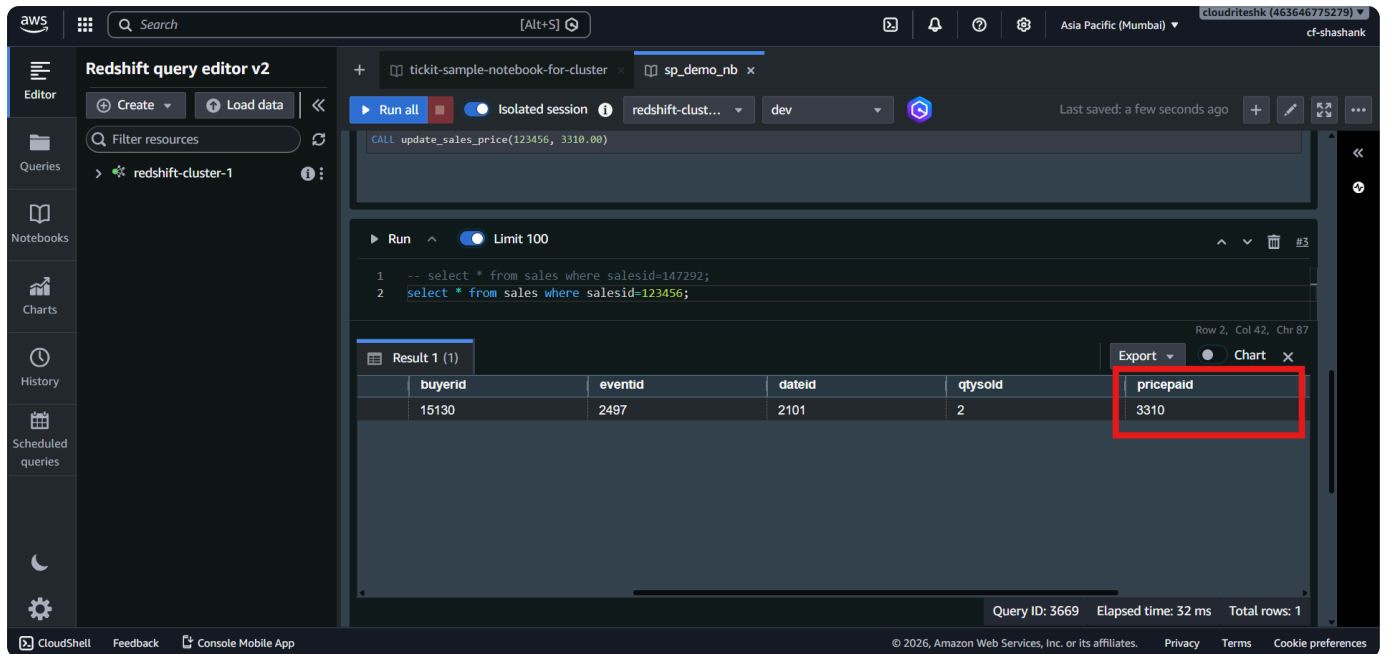
Price passed: 100

Result

Received expected notice: No sales record found for provided sales ID

Test 2 – Valid Sales ID





Valid sales ID used: 123456

Existing value: 331

Updated value: 3310

Procedure executed successfully.

Validation Query

Ran query again on: salesid = 123456

Result

Verified: pricepaid = 3310

The update was successfully applied.

Key Learning

Stored Procedures in Redshift help:

- Automate SQL operations
- Reuse database logic
- Handle validations
- Perform complex data warehouse processing

Useful for:

- Batch updates
- ETL workflows
- Data validation
- Business rules